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WEARABLE AI FOR EARLY DETECTION OF PARKINSON'S & ALZHEIMER'S DISEASE

DEPARTMENT - CSE(AIML)

YEAR - 4TH YEAR

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INTRODUCTION

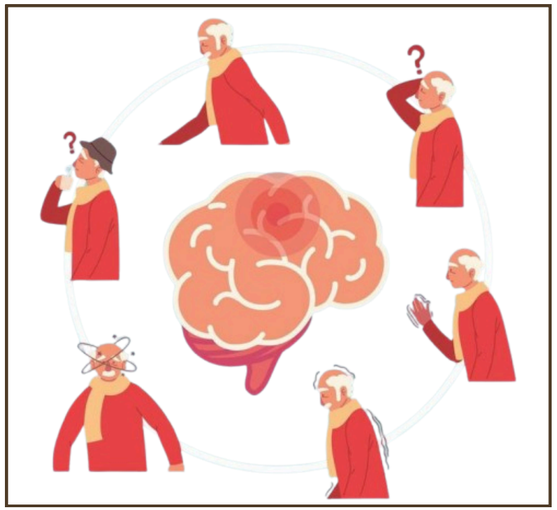
“Alzheimer’s and Parkinson’s disease are progressive neurodegenerative disorders that affect millions of people worldwide.”

- **Alzheimer’s** causes memory loss and cognitive decline, while **Parkinson’s** affects movement, leading to tremors, stiffness, and balance issues. Both diseases worsen over time and often go undiagnosed until significant neurological damage has occurred.

- **Early detection** is crucial as it allows for **timely medical intervention**, **slows disease progression**, and **improves quality of life**. It helps families **plan for long-term care**. However, traditional diagnostic methods rely on clinical observations and tests that may not detect symptoms early enough.

- Our AI-powered watch detects early signs of Alzheimer’s and Parkinson’s through **motion tracking**, **speech analysis**, and a **task scheduler**.

- It enhances **safety with GPS**, **emergency alerts**, and aids symptom relief via **neurostimulation therapy**, providing real-time health insights for patients, caregivers, and doctors.



HOW OUR WEARABLE AI WORKS

Data Collection & Monitoring

- **Motion Sensors**: Detect tremors and gait abnormalities.
- **Speech Analysis**: Tracks voice modulation, hesitation, and slurring.
- **Heart Rate & Stress Sensors**: Monitor nervous system irregularities and emotional distress.

Connectivity & Data Sharing

- **Cloud Storage & AI Reports**: Securely stores data and tracks disease progression over time.
- **Caregiver & Doctor Access:** Enables real-time health monitoring and remote intervention.
- **Custom Alerts:** Notifies users and caregivers of critical health trends.

Continuous Learning & Adaptation

- **AI Model Updates**: Enhances detection accuracy.
- **Adaptive Monitoring**: Customizes tracking by disease stage.
- **User Feedback Integration**: Refines usability & alerts.

AI-Driven Analysis

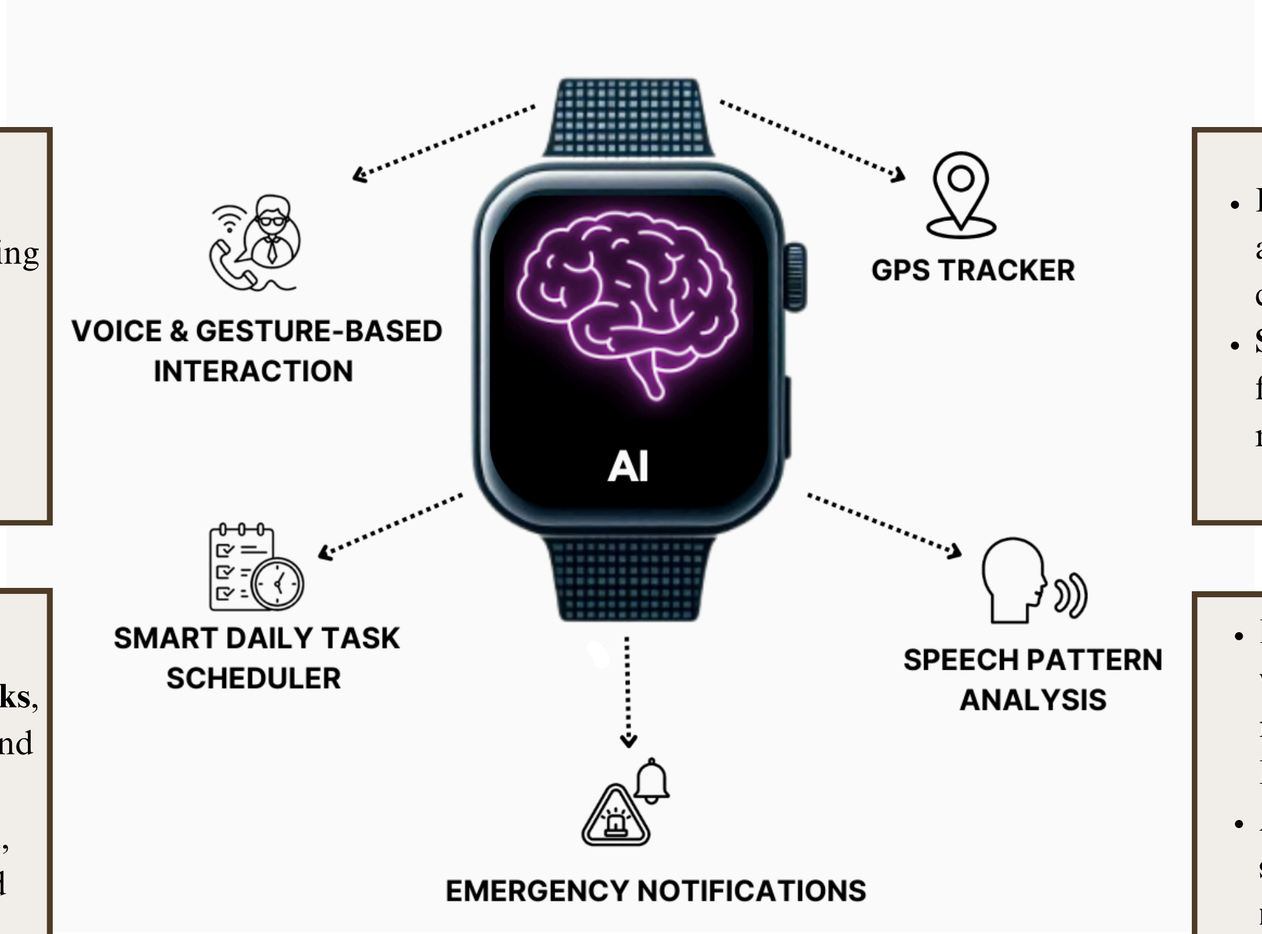
- **Machine Learning**: Analyzes movement and speech for early disease markers.
- **Anomaly Detection**: Identifies behavioral deviations and alerts caregivers.
- **Predictive Insights**: Warns of potential neurological decline based on trends.

Smart Assistance

- **Emergency Alerts & GPS Tracking:** Detects falls, prevents wandering.
- **Task Reminders:** Assists with schedules daily routines.
- **Voice Interaction:** Enables hands-free navigation for easier usability.

INNOVATIVE ENHANCEMENTS

- Detects speech changes like **hesitation**, and **slurring**, indicating **cognitive** and **motor decline**.
- Gesture controls help patients with **tremors** for smoother **interactions**.



- Recognizes **wandering behavior** or abrupt stops, common in both conditions.
- **Sends alerts** if the patient strays from familiar locations or experiences mobility difficulties.

- Monitors missed **medications**, **appointments**, or **incomplete tasks**, **detecting memory impairment** and **motor difficulties.**
- Tracks delays in task execution, indicating cognitive decline and reduced motor control.

- Detects **speech hesitation**, **monotone voice**, and **slowed speech**, early markers of **Alzheimer's** and **Parkinson's**.
- Analyzes changes in **vocabulary**, **sentence structure**, and **voice modulation** to assess **neurological health.**

- Identifies **sudden falls**, **tremors**, and **changes in movement patterns**, signaling motor impairments and cognitive decline.
- Detects **restlessness**, **disorientation**, or **slow movements**, **triggering** alerts for caregivers.

TECHNOLOGY STACK

Hardware Components

- **Biometric Sensors**: Track heart rate, temperature, and oxygen levels for physiological changes.
- **Motion Sensors**: Monitor gait, balance, and tremors using accelerometers & gyroscopes.
- **Speech Processing Units**: Analyze voice patterns for cognitive and speech impairments.
- **GPS Module**: Enables real-time location tracking and safety alerts.
- **Edge AI Chips**: Process sensor data in real-time for faster insights without cloud reliance.

Software & AI models

- **Machine Learning Frameworks:** TensorFlow, PyTorch for detecting speech irregularities and movement patterns.
- **Neural Networks:** LSTM & CNN models for analyzing cognitive function using speech and motor data.
- **Cloud Computing:** AWS, Google Cloud, Firebase for secure data storage and remote monitoring.
- **Mobile & Web Applications:** Flutter for cross-platform mobile apps, React for web dashboard interfaces.
- **Embedded Software for Wearable Device:** Arduino, Raspberry Pi, ESP32 for on-device AI processing in real-time.

ETHICAL CONSIDERATIONS & SOLUTIONS

Data Privacy & Security
- Sensitive Health Data
- End-to-End Encryption
- User Control

Reliability & Accuracy
- Medical-Grade Precision
- Continuous Model Updates

Informed Consent
- Transparency
- Opt-in Mechanism
- Easy Opt-Out

Autonomy & Human Oversight
- AI as an Aid, Not a Replacement
- User Empowerment

Bias & Fairness in AI
- Diverse Training Data
- Bias Mitigation

Ethical AI Deployment & Regulation
- Compliance with Health Regulations
- Independent Audits

Accessibility & Inclusivity
- Affordable Solutions
- User-Friendly Design

SOCIETAL IMPACT

Early Detection & Prevention

- Identifies Alzheimer's & Parkinson's symptoms before they worsen.
- Helps doctors start treatment sooner, improving patient outcomes.
- Reduces long-term healthcare costs by delaying disease progression.

Improved Quality of Life

- Supports independent living with smart assistance.
- Reduces stress by offering continuous health monitoring.
- Enhances daily functioning with reminders and symptom management.

Advancing Medical Research

- Generates real-world health data for studying neurodegenerative diseases.
- Helps researchers improve early diagnosis techniques.
- Supports AI-driven medical advancements in brain health.

Raising Awareness & Reducing Stigma

- Encourages early screening and proactive health management.
- Educates the public on the importance of brain health.
- Reduces fear around Alzheimer's & Parkinson's.

Caregiver & Family Support

- Provides real-time health updates to caregivers and doctors.
- Reduces the burden of constant supervision with automated alerts.
- Increases safety with fall detection and GPS tracking

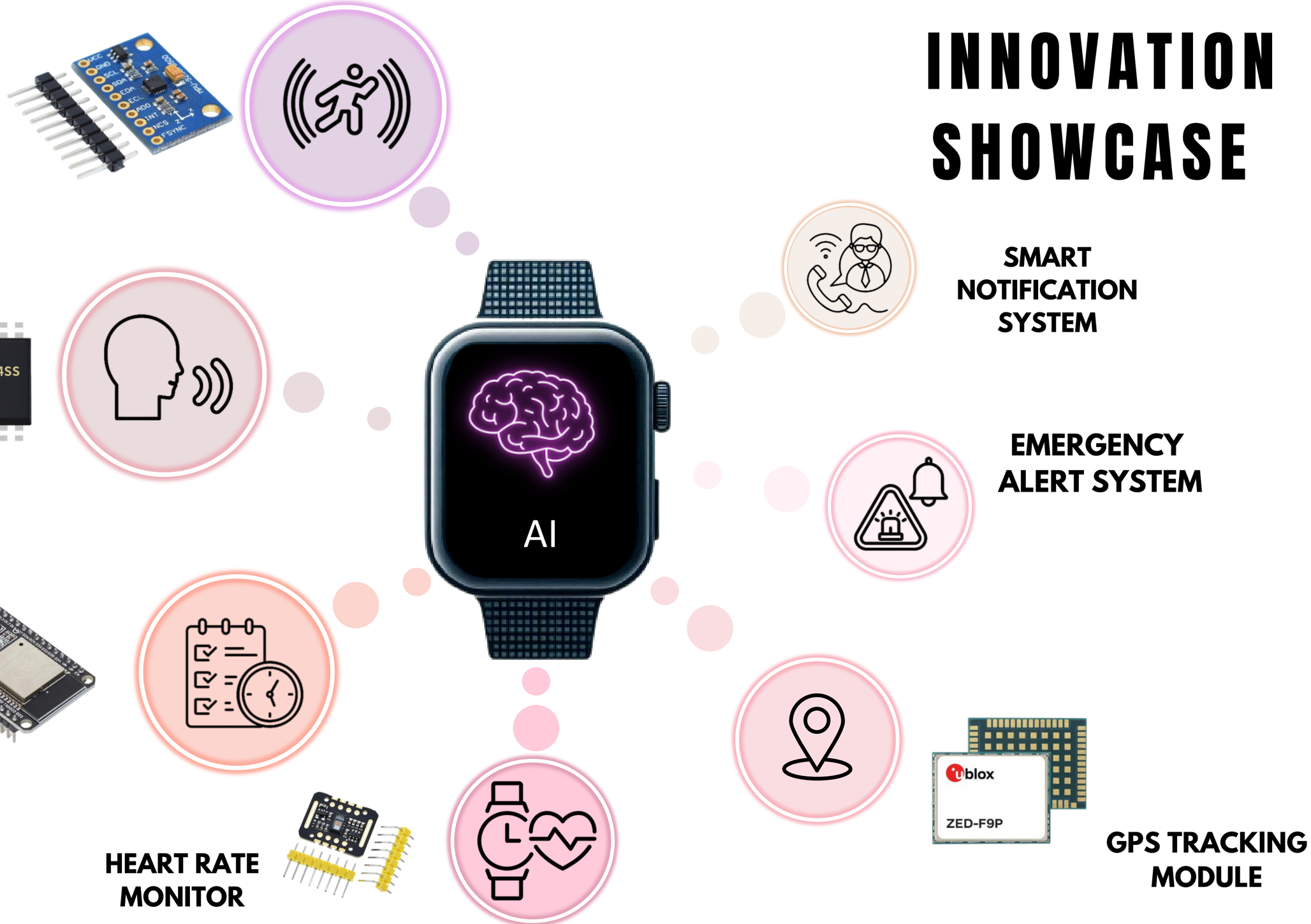
Affordable & Accessible Healthcare

- Offers a cost-effective solution for disease monitoring.
- Expands access to healthcare, especially in underserved areas.
- Reduces hospital visits through home-based monitoring.

CONCLUSION

- **Wearable AI** revolutionizes early detection and proactive management of Alzheimer’s & Parkinson’, enabling timely intervention.
- Our **AI-powered innovations** ensure **real-time monitoring**, enhanced safety, and personalized assistance, improving patients' quality of life.
- By integrating **ethical safeguards**, including **data privacy** and user consent, we ensure a responsible, transparent, and impactful healthcare solution.
- This **cutting-edge technology** not only empowers individuals and caregivers but also advances medical research for better neurodegenerative disease management.

INNOVATION SHOWCASE



Wearable AI for Early Detection of Parkinson’s & Alzheimer’s Disease

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ABSTRACT

Neurodegenerative diseases like Alzheimer’s and Parkinson’s often go undetected until advanced stages, limiting treatment options.

This AI-powered wearable continuously monitors motor function, speech patterns, and biometrics to identify early signs of these conditions.

Equipped with motion sensors, speech analysis, and GPS tracking, it provides real-time health insights and assists with daily tasks.

Machine learning algorithms analyze collected data, enabling early intervention and remote monitoring. This innovation enhances patient care, supports caregivers, and advances proactive healthcare solutions.

INTRODUCTION

“Alzheimer’s and Parkinson’s disease are progressive neurodegenerative disorders that affect millions of people worldwide.”

Alzheimer’s causes **memory loss** and **cognitive decline**, while Parkinson’s **affects movement**, leading to **tremors**, **stiffness**, and **balance issues**. Both diseases worsen over time and often go undiagnosed until significant neurological damage has occurred.

Early detection is crucial as it allows for timely medical intervention, slows disease progression, and improves quality of life. It helps families plan for long-term care. However, traditional diagnostic methods rely on clinical observations and tests that may not detect symptoms early enough. Our AI-powered wearable detects early signs of Alzheimer’s and Parkinson’s through **motion tracking**, **speech analysis**, and a **task scheduler**.

It enhances safety with **GPS**, **emergency alerts**, and aids symptom relief via neurostimulation therapy, providing real-time health insights for patients, caregivers, and doctors.

METHODS AND MATERIALS

The **hardware components** include biometric sensors, motion sensors, speech processing units, GPS module, neurostimulation unit, and edge AI chips. The **software and AI models** utilize **TensorFlow**, **PyTorch**, LSTM**,** CNN, AWS, Google Cloud, **Firebase**, **Flutter**, **React**, **Arduino**, Raspberry Pi, and ESP32.
Our **methodology** involves **real-time data collection** from motion sensors, microphones, and biometric trackers to monitor **speech**, **movement**, and **cognitive patterns**. AI algorithms analyze this data to detect early symptoms, identify abnormalities, and generate alerts. A **cloud-based system** ensures secure storage and remote monitoring, enabling timely intervention through a connected **mobile app** for patients and caregivers.

Figure 1. Proposed Wearable Prototype

Figure 2. Technology Stack

RESULTS

Wearable AI devices revolutionize the early detection and management of neurodegenerative diseases like Parkinson's and Alzheimer's. By providing **continuous**, **real-time monitoring** and **personalized health insights**, these devices detect early cognitive and motor decline. This particular device aims to identify Parkinson's and Alzheimer's symptoms, such as **subtle tremors**, **gait changes**, **speech irregularities** and **memory lapses**.

It will also capture early motor fluctuations by 85-90% sensitivity and a 25-35% improvement in Alzheimer's diagnosis accuracy, with significantly enhancing detection capabilities. The **continuous tracking** will also ensure that **episodic** and **mild symptoms**, often missed in clinical visits, are flagged for **timely intervention**. Real-time alerts improve safety and **emergency response**, while long-term monitoring aids in accurate diagnosis and trend analysis.

It will make sure that almost 70% of participants will experience **slower disease progression** and **early detection**. By seamlessly integrating with healthcare systems, this device will support data-driven research and enable more precise, proactive care, ultimately enhancing well-being and providing a more effective, preventive approach to managing neurodegenerative conditions.

PATIENT ID	AGE	GENDER	TREMORS (HZ)	GAIT SPEED (M/S)	SPEECH PAUSE DURATION (MS)	MEMORY TEST SCORE	MRI BRAIN SCAN (LESION %)	DIAGNOSIS
P001	68	Male	2.5	0.8	750	40	12	Alzheimer's
P002	72	Female	5.6	0.5	620	78	6	Parkinson's
P003	65	Male	1.8	1.1	780	35	15	Alzheimer's
P004	70	Female	6.2	0.4	590	82	5	Parkinson's
P005	75	Male	2.2	0.7	730	50	10	Alzheimer's

Table 1. Alzheimer's & Parkinson's Patient Data

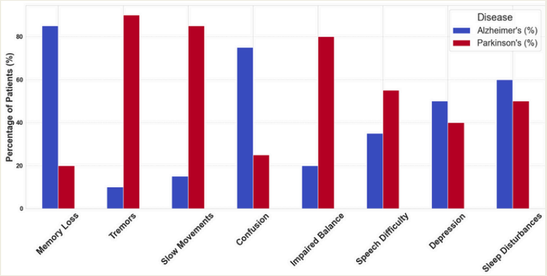

Chart 1. Comparison of Symptoms in Alzheimer's & Parkinson's Patients

DISCUSSION

The wearable device is designed to detect symptoms of cognitive and motor decline, enabling timely interventions and support. It tracks speech changes, such as hesitation, slurring, and monotone voice, which can indicate **cognitive impairment**. The device also monitors **movement patterns**, detecting tremors, sudden falls, and changes in gait or balance. Additionally, it recognizes wandering behavior, restlessness, and **disorientation**, common in individuals with **cognitive decline**. The device sends **alerts to caregivers** if the patient **strays** from **familiar locations**, experiences **mobility difficulties**, or exhibits other **concerning behaviors**. It also analyzes changes in vocabulary, sentence structure, and voice modulation to assess neurological health. By providing **real-time insights** into cognitive and motor function, the wearable device facilitates prompt care and support, enabling individuals to maintain their **independence** and **quality of life**. Caregivers can respond quickly to alerts, ensuring the patient receives the help they need.

CONCLUSIONS

Wearable AI revolutionizes early detection and proactive management of Alzheimer’s & Parkinson’, **enabling timely intervention**.
Our AI-powered innovations ensure **real-time monitoring**, **enhanced safety**, and **personalized assistance**, improving patients' quality of life.
By integrating ethical safeguards, including **data privacy** and user consent, we ensure a responsible, transparent, and impactful healthcare solution.
This **cutting-edge technology** not only empowers individuals and caregivers but also advances **medical research** for better neurodegenerative disease management.

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